

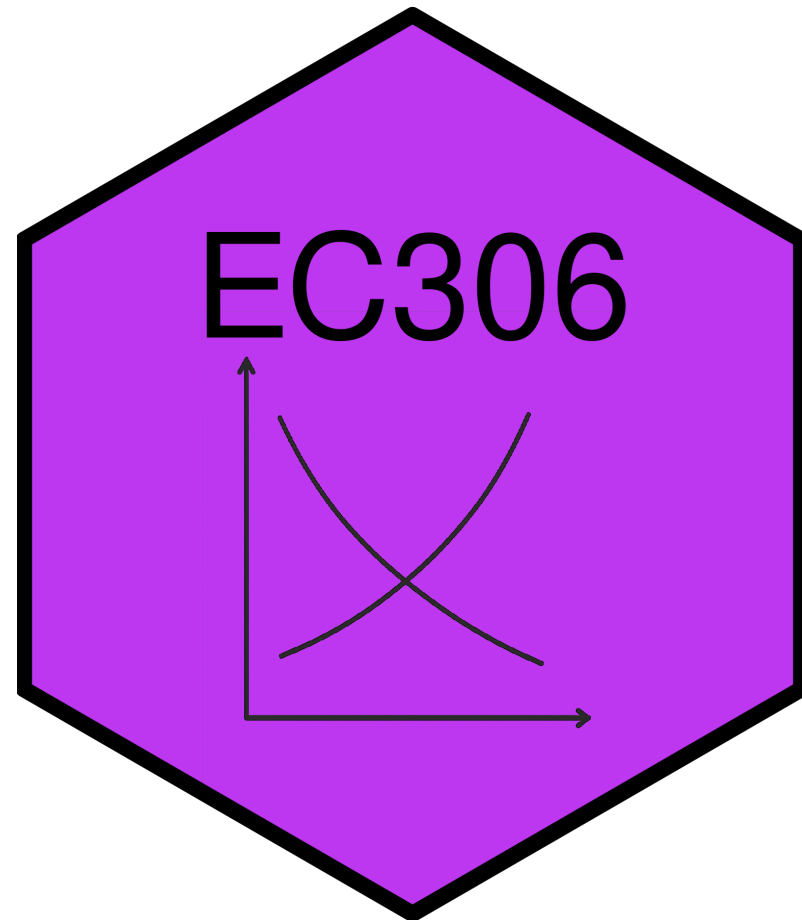
Labour Demand in Global Markets

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Understand the impact of free trade and international competition on the Canadian labour market
- Analyze the effect of trade on a single labour market
- Consider the cross-sectoral impact of trade
- Examine the evidence on trade and labour demand

Globalization and Offshoring

Introduction

Over time Canadian firms have faced increasing globalization

Has led to concerns over what happens to Canadian workers:

- People worry firms will move operations to countries with lower wages
- That this will destroy Canadian jobs
- Clear concern in the USA right now

Note

Effect of globalization on jobs is always a concern when trade agreements are reached — manufacturing is especially prominent, with automotive industry currently a focus

Effect of Trade on Single Market

Consider global competition with a specific product

Increased global competition lowers the price a product can sell for globally

In a competitive market, this reduces the value of the marginal product:

- Recall that $VMP = p \times MP_N$
- Shifts demand for Canadian labour inward

Effect of Trade on Single Market

Another approach: consider Canadian and foreign labour as two inputs to production

The optimal choice of Canadian and foreign labour occurs where:

$$\frac{MP_{N_C}}{MP_{N_F}} = \frac{w_C}{w_F}$$

The marginal product per dollar spent on foreign and Canadian workers are the same

Effect of Trade on Single Market

What effect does a fall in foreign wages have on Canadian labour demand?

Two effects occur:

- **Substitution effect:** firms substitute foreign workers for Canadian workers
 - Reduces demand for Canadian workers
- **Scale effect:** lower costs lead to more production
 - Increases demand for Canadian workers

Overall Effect is Ambiguous

- If close substitutes, substitution effect dominates → less Canadian labour demanded
- If scale effect dominates → could be more Canadian labour demanded

Effect of Trade on Single Market

! Cost is Not the Only Factor

Productivity is also important — if Canadian workers are more productive, firms will still want to hire them

How productive are Canadian workers compared to foreign workers?

Next slides show international comparisons for OECD countries:

- Productivity, compensation, unit labour costs
- Changes over time

Effect of Trade on Single Market

TABLE 9.1 Productivity¹ in 2010 and Changes in Productivity, Hourly Compensation,² and Unit Labour Costs³ between 2010–2022 (select OECD countries)

Country	Productivity in 2010 (as % of Canada)	% Change in Productivity	% Change in Compensation	% Change in Unit Labour Costs
Norway	161	6	-9	-14
Belgium	156	6	0	-6
Denmark	148	13	-1	-12
Netherlands	144	7	-1	-7
France	144	5	-5	-9
Switzerland	140	13	24	10
Sweden	138	10	0	-13
Germany	138	12	10	-2
United States	135	12	36	20
Finland	130	4	-3	-8
United Kingdom	127	6	3	-3
Italy	124	4	-11	-14
Spain	104	9	-7	-14
Canada	100	12	9	-1
New Zealand	84	10	26	13
Japan	84	11	-25	-32
Korea	70	33	39	6
Mexico	41	-4	8	10
Average	120	9	5	-4

Notes:

1. GDP per hour worked (constant US dollars, PPP converted). Percentage of Canadian value, ranked in descending order.

2. Sum of gross wages and salaries and employers' social security contributions divided by total hours worked by employees. All figures are on an hours-worked basis and are converted to Canadian dollars and adjusted for inflation using the Canadian annual CPI (all-items) series.

3. Average cost of labour per unit of output produced. All figures are converted to Canadian dollars and adjusted for inflation using the Canadian annual CPI (all-items) series.

Source: Authors' calculations based on OECD data. Available online at <https://www.oecd.org/en/data/datasets/oecd-DE.html>.



Effect of Trade on Single Market

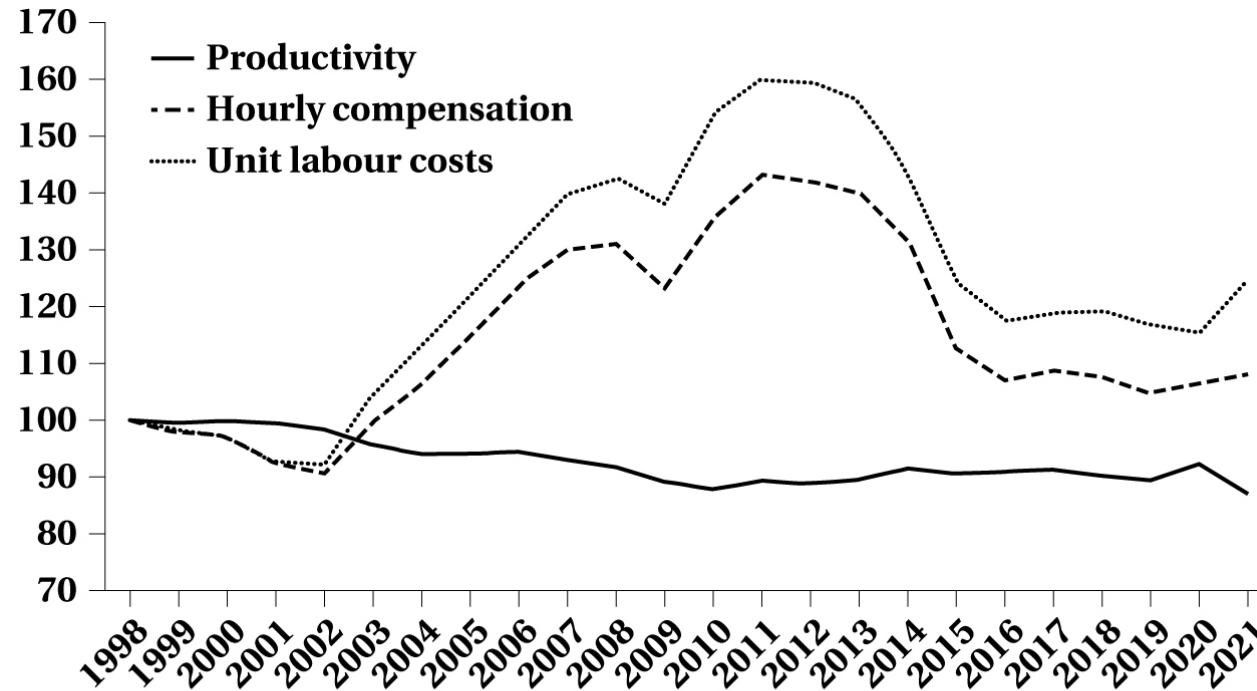


Figure 9.1 Change in Canadian Productivity, Hourly Compensation, and Unit Labour Costs Relative to the United States (1998 = 100)

This figure plots the change since 1998 in Canadian productivity, hourly compensation, and unit labour costs as a percentage of United States. Each value is the ratio of the Canadian and US index where each country's series is 100 in 1998. The unit labour costs series tells us that by 2015, real hourly compensation expressed in Canadian dollars had grown 25 percent more in Canada than in the United States. The figures do not compare the level of productivity and costs between Canada and the United States; only their relative changes since 1998.

Note: Values represent the ratio of the Canadian and US index of productivity and labour costs (1998 = 100).

Source: Authors' calculations based on OECD data. Available online at <https://www.oecd.org/en/data/datasets/oecd-DE.html>.

Effect of Trade Across Sectors

Single Sector Misses Key Points

Trade openness affects the entire economy with adjustments across sectors

Recall from EC120: countries specialize in production of goods where they have comparative advantage

- Expands consumption possibilities

In a globalized market:

- Countries allocate more resources towards their comparative advantage sectors
- Labour moves from non-comparative advantage sectors to comparative advantage sectors

Evidence on Trade and Labour Demand

Many trade agreements with other countries

Most prominent in Canada: NAFTA and more recently CUSMA

These present opportunities to study effects of trade on labour demand

Evidence Shows

- About 10% of Canadian jobs lost in early 90s were due to free trade
- Productivity grew faster in manufacturing industries affected by trade

Non-Trade Demand Factors

At the same time trade is changing, the Canadian labour market is changing:

- Jobs have shifted away from manufacturing over long term
- Professional, managerial, and administrative service jobs have been increasing
- Flex or “gig” jobs have been increasing

All changing for various reasons:

- Technology
- Consumer preferences
- Demographics

Think About It

Suppose a Canadian auto parts manufacturer faces a drop in wages in Mexico. Under what conditions would this actually *increase* demand for Canadian auto workers rather than decrease it?

Summary

Summary

Trade and labour demand:

- Trade shifts labour demand, but the direction differs by sector
- A single-market analysis misses the reallocation across sectors
- Cost is not the only factor driving offshoring decisions
- The evidence finds effects concentrated in trade-exposed industries